

MolGuidance: Advanced Guidance Strategies for Conditional Molecular Generation with Flow Matching

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S-1 Mathematical Formulation of Discrete Guidance Formats

Linear Interpolation (Additive Blend) Take guidance on probability distribution as an example.

$$p_{\text{guided}} = (1 - w) p_{\text{unconditional}} + w p_{\text{conditional}} \quad (1)$$

This formula represents a direct, weighted average of the two probability distributions.

Logarithmic Interpolation (Multiplicative Blend)

$$p_{\text{guided}} = \exp((1 - w) \log p_{\text{unconditional}} + w \log p_{\text{conditional}}) \quad (2)$$

$$= \exp(\log p_{\text{unconditional}}^{1-w} + \log p_{\text{conditional}}^w) \quad (3)$$

$$= \exp(\log(p_{\text{unconditional}}^{1-w} \cdot p_{\text{conditional}}^w)) \quad (4)$$

$$= p_{\text{unconditional}}^{1-w} \cdot p_{\text{conditional}}^w \quad (5)$$

This is the form operating in log-space, which is mathematically equivalent to a multiplicative blend.

S-2 Interpolants, Priors and Loss Functions

Flow for atomic positions takes a linear interpolant:

$$X_t = (1 - t)X_0 + tX_1 \quad (6)$$

where X_0 and X_1 are initial and final states. The base distribution (prior) for atomic positions use a centered standard normal distribution $p_0(X) = \prod_{i=1}^N \mathcal{N}(X_0^i | \mathbf{0}, \mathbb{I}_3)$. The optimization of the conditional velocity field for atomic positions can be reparameterized into the optimization of a denoiser network with an mean squared error (MSE) objective:

$$\mathcal{L}_X = \mathbb{E}_{t, p_t(X_t | X_0, X_1), \pi(X_0, X_1)} [\|X_{1|t}^\theta - X_1\|] \quad (7)$$

Where the joint distribution $\pi(X_0, X_1)$ defines the optimal transport coupling between (X_0, X_1) . $X_{1|t}^\theta$ represents the predicted atomic position given the state at time t . Details of the optimal transport formulation are provided in section S-4.

Discrete variables such as atom types and charges are modeled through the CTMC flows [1]. The prior distribution is the state in which all atoms are in a masked state, and the generation process is essentially a demasking process. We refer the readers to the FlowMol work for more details of the CTMC flows [2]. The objective for these discrete variables takes the cross-entropy format:

$$\mathcal{L}_{\text{CE}} = \mathbb{E}_{t, p_{t|1}(x_t | z), p_z} [-\log p_{t|1}^\theta(x_t^i | x_t)] \quad (8)$$

The total loss for the molecule graph is a weighted linear summation of losses for each molecular modality:

$$\mathcal{L} = \eta_X \mathcal{L}_X + \eta_A \mathcal{L}_A + \eta_C \mathcal{L}_C + \eta_E \mathcal{L}_E \quad (9)$$

Empirically, it is preferential to determin atomic positions first, followed by bonds, charges and atom types. In view of this, the loss weights are chosen to be $(\eta_X, \eta_A, \eta_C, \eta_E) = (3.0, 0.4, 1.0, 2.0)$.

S-3 Model Architecture

The vanilla PropMolFlow model uses the ‘Concatenate_Sum’ property embedding method without a Gaussian expansion, as implemented in our previous work [3]. The backbone model for SE(3)-equivariant joint flow matching adopts the architecture implemented in FlowMol [4]. Molecule updates are achieved through layers comprising of Geometric Vector Perceptrons (GVPs). Within each GVP, the molecule graph passes through a sequential steps of Node Feature Update (NFU), Node Position Update (NPU) and Edge Feature Update (EFU). Each node i consists of a position $x_i \in \mathbb{R}^3$, scalar features $s_i \in \mathbb{R}^d$, and vector features $v_i \in \mathbb{R}^{c \times 3}$. Non-zero vector features are involved a cross-product vector operation, which is crucial to break the reflection symmetry, making it an SE(3) equivariant architecture. The scalar feature is a concatenation of atom type and charge vectors; that is, $s_i = [a_i \oplus c_i]$ where ‘ $\oplus$ ’ defines a concatenation operation. Each edge feature corresponds to the bond order, and the permutation invariance of bond orders is realized through taking the sum of bond features from $i \rightarrow j$ and $j \rightarrow i$; that is, $\hat{e}^{ij} = \text{MLP}(e_{ij} + e_{ji})$.

Node Feature Update. The node feature takes two steps for its update: the first step generates scalar messages $m_{i \rightarrow j}^{(s)}$, and vector messages $m_{i \rightarrow j}^{(v)}$ by a function ψ_M which corresponds to a sequential two GVPs:

$$m_{i \rightarrow j}^{(s)}, m_{i \rightarrow j}^{(v)} = \psi_M \left(\left[s_i^{(l)} \oplus e_{ij}^{(l)} \oplus d_{ij}^{(l)} \right], \left[v_i \oplus \frac{x_i^{(l)} - x_j^{(l)}}{d_{ij}^{(l)}} \right] \right) \quad (10)$$

Where $d_{ij}^{(l)}$ is the distance between nodes i and j at the update layer l . To enrich the neighboring environment, the distance d_{ij} is expanded with a radial basis function (RBF) embedding before fed into the next GVPs or MLPs. A message passing procedure is conducted to update node scalar and vector features by aggregation:

$$s_i^{(l+1)}, v_i^{(l+1)} = \text{LN} \left(\left[s_i^{(l)}, v_i^{(l)} \right] + \psi_N \left(\frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} \right) \left[m_{j \rightarrow i}^{(s)}, m_{j \rightarrow i}^{(v)} \right] \right) \quad (11)$$

Where LN stands for s LayerNormalization operation [5], ψ_N is a chain of three GVPs.

Node Position Update. Node-wise operations are used on the updated node scalar and vector features to update node position features:

$$x_i^{(l+1)} = x_i^{(l)} + \psi_P \left(s_i^{(l+1)}, v_i^{(l+1)} \right) \quad (12)$$

Where ψ_P is a sequential three GVPs in which the final output has 1 vector and 0 scalar features.

Edge Feature Update. Edge features are updated by edge-wise operations that takes the updated node scalar features and node distance as the inputs:

$$e_{ij}^{(l+1)} = \text{LayerNorm} \left(e_{ij}^{(l)} + \text{MLP} \left(s_i^{(l+1)}, s_j^{(l+1)}, d_{ij}^{(l+1)} \right) \right) \quad (13)$$

S-4 Equivariant Optimal Transport

To smooth the probability path, we used the equivariant optimal transport to align the initially sampled noisy molecules with the target molecules, as used in previous works [6–8]. This is achieved by optimal permutation of atom node indices and rigid body alignment between base molecules and target molecules, for both of which their center-of-masses are removed to respect the translational invariance [8, 9].

S-5 Additional Results

S-5.1 Different Discrete Guidance Result for 5 Property in QM9 Dataset

Result is shown in Figure S1.

S-5.2 Hybrid Guidance Result for 5 Property in QM9 Dataset

Result is shown in Figure S2.

S-5.3 Bayesian optimized guidance weights targeting property alignment.

Figure S3 illustrates a schematic example of Bayesian optimization (a), the results for CFG weights optimized over the HOMO property (b), AG weights optimized over heat capacity (c), and MG single weight optimized over HOMO-LUMO gaps (d).

MAEs for the MG models vary in a smaller range, suggesting the difficulty in leveraging the guidance weight effects for MG. Also, there are high fluctuations for MAE when guidance weights changes from around 1.6 to 1.7, indicating the instability of the MG model for describing the guidance weight effect.

Table S1: Bayesian optimized guidance weights (w_1, w_2) for each method in two datasets. For MG, identical weights apply to both positional and discrete variables. All weights are dimensionless.

Property	α	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	$\mu\text{-QM9}$	C_v	$\mu\text{-QMe14S}$
PMF-CFG	(4.00, 1.77)	(4.00, 2.16)	(2.71, 1.91)	(3.97, 2.29)	(4.00, 2.31)	(4.00, 2.00)	(3.13, 1.88)
PMF-AG	(2.34, 1.00)	(4.26, 1.34)	(2.79, 1.11)	(3.15, 1.17)	(4.29, 1.50)	(2.75, 1.14)	(2.71, 1.11)
PMF-MG	1.34	1.68	1.89	2.23	2.21	2.14	2.25

It is crucial to identify the optimal guidance as the MAEs can vary by more than ten times in certain situations (e.g., from 1.40 to 15.5 Bohr³ for AG conditioned on α). The guide model that yields the best performance under AG is provided in Table S3. As a general rule, greater differences between the guide and main models tend to produce lower property MAEs. One may argue that expanding the search range can further improvement the property alignment, since many optimal atomic-position weights (w_1) lie at the boundaries of the search range for both CFG and AG (Table S1). Nevertheless, given the weaker MAE dependence on w_1 , the additional gain by extending the search space may not be worthwhile. All subsequent performance metrics are computed using models with their corresponding best guidance weights.

Ranges of MAEs for Bayesian optimization can be found in Table S2.

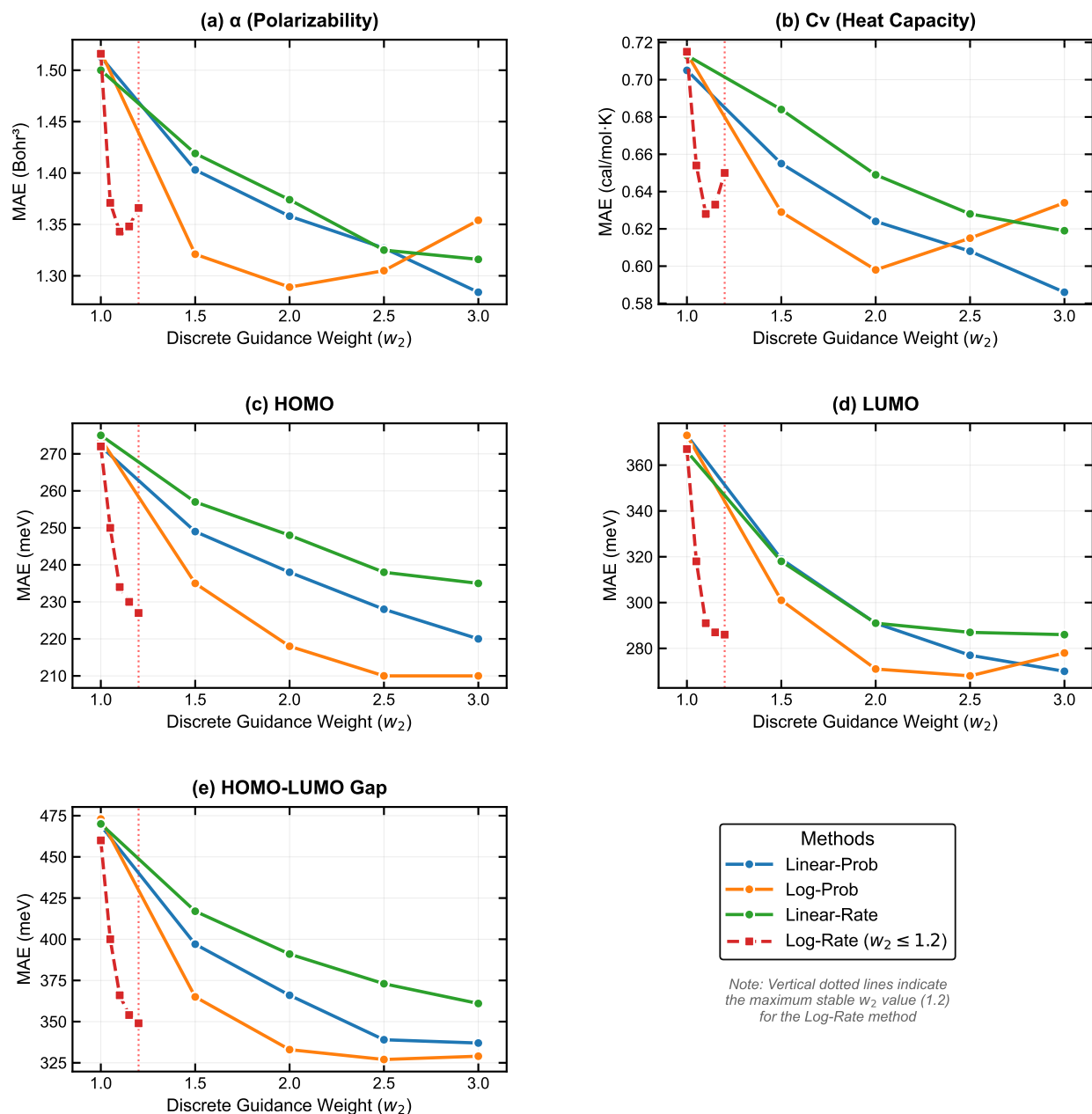


Figure S1: Comparison of discrete guidance formats for qm9 dataset across 5 properties. Four guidance methods are compared: linear guidance on probability (Linear-Prob), logarithm guidance on probability (Log-Prob), linear guidance on rate matrix (Linear-Rate), and logarithm guidance on rate matrix (Log-Rate). The Log-Rate method is constrained to $w_2 \leq 1.2$ due to numerical instability at higher guidance weights, indicated by the vertical dotted line. Lower MAE values indicate better performance.

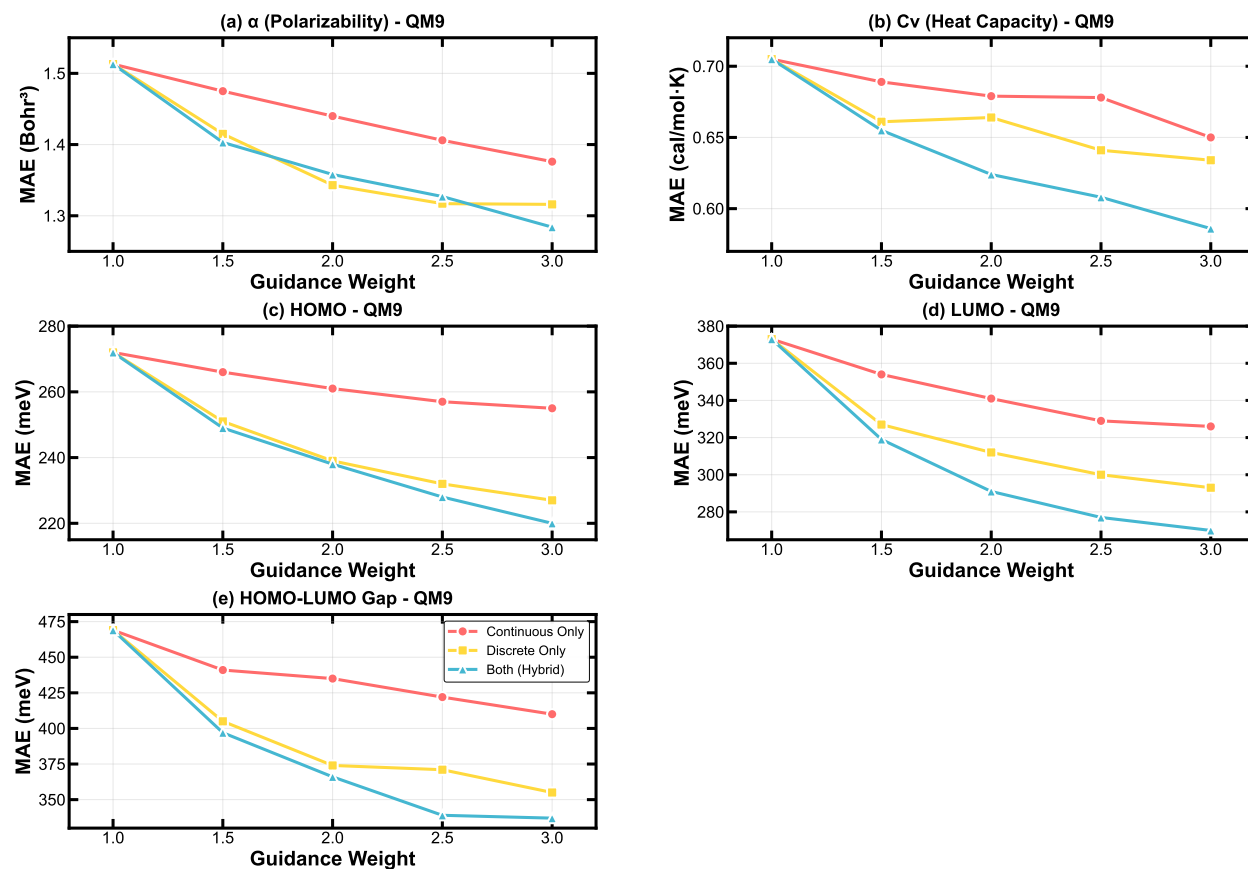


Figure S2: MAE for 5 properties alignment using three guidance strategies: continuous-only (atomic positions), discrete-only (atom types, charges, bonds), and hybrid (both domains) on QM9 datasets. Lower MAE values indicate better alignment between target and generated molecule properties

Table S2: Ranges of MAEs for Bayesian optimization for different guidance methods. Results are evaluated on 1000 sampled molecules. AG uses the guide model trained with 40000 steps and reduced numbers of node and edge features.

Property Units	α Bohr ³	$\Delta\epsilon$ meV	ϵ_{HOMO} meV	ϵ_{LUMO} meV	μ -QM9 Debye	C_v cal/(mol·K)	μ -QMe14S
CFG, Minimum MAE	1.20	317	202	249	0.549	0.556	0.508
CFG, Maximum MAE	2.68	603	282	757	0.746	1.684	0.708
AG, Minimum MAE	1.40	328	231	256	0.591	0.616	0.560
AG, Maximum MAE	15.5	425	360	466	0.926	1.023	0.679
MG, Minimum MAE	1.53	406	261	333	0.714	0.689	0.628
MG, Maximum MAE	1.67	457	280	358	0.778	0.759	0.704

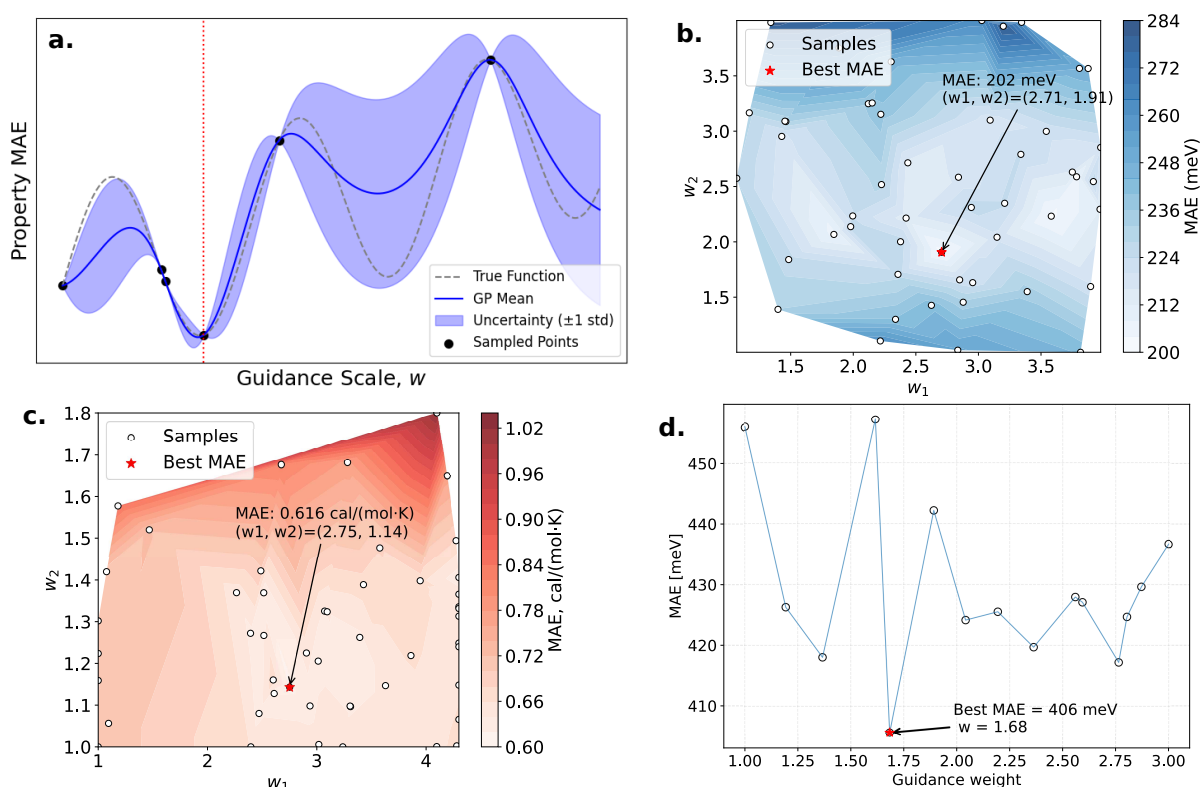


Figure S3: Bayesian optimization of guidance weights for property alignment. (a) A 1-D schematic plot of the Bayesian optimization aiming to minimize the property MAEs. (b) Performance of Bayesian optimization over guidance weights for CFG conditioning on HOMO energy. (c) Bayesian optimization over guidance weights for AG conditioning on heat capacity. (d) Bayesian optimization over single guidance weights for MG conditioning on HOMO-LUMO gap values. Hollow circles denote sampled weight pairs, with the best candidate (lowest MAE) highlighted by a red star; its guidance weights and MAE are indicated. MAEs were computed over 1000 molecules sampled from the joint distribution of atom count and property values.

Comparing performance of two guide models in AG. The best guide model for each property is summarized in Table S3. The first guide model ($u_{g,1}$) is trained with 40000 steps and has a reduced-by-half numbers of hidden node features and edge features, and the second guide model ($u_{g,2}$) is the same architecture of the main model but trained with only 51 epochs.

Table S3: Bayesian optimized guidance weights and the best MAEs for two different guide models in AG for the QM9 data. Numbers in bracket correspond to (w_1, w_2) , which are dimensionless, and values after the weights are corresponding best MAEs. This evaluation is based on 1000 sampled molecules, and the model chosen for further analysis in the main text are **bold**.

Property Units	α Bohr ³	$\Delta\epsilon$ meV	ϵ_{HOMO} meV	ϵ_{LUMO} meV	μ Debye	C_v cal/(mol·K)
$u_{g,1}$	(2.09, 1.21), 1.40	(4.26, 1.34), 328	(2.79, 1.11), 231	(3.15, 1.17), 256	(4.29, 1.50), 0.591	(2.75, 1.14), 0.616
$u_{g,2}$	(2.34, 1.00), 1.39	(2.47, 1.42), 330	(1.93, 1.28), 234	(3.63, 1.15), 272	(4.29, 1.20), 0.631	(1.00, 1.35), 0.672

S-5.4 Effect of Guidance Weights on Structural Validity and Diversity

Table S4–S6 report how validity, molecule stability, uniqueness, and novelty vary with guidance weights for the QMe14S dataset conditioned on μ . Three guidance configurations are examined: hybrid guidance (both w_1 and w_2 varied simultaneously), continuous-only guidance (varying w_1 with $w_2 = 1.0$ fixed), and discrete-only guidance (varying w_2 with $w_1 = 1.0$ fixed). The validity and molecular stability of the generated molecules are evaluated using the method proposed by Nikitin et al. [10]. Uniqueness is measured as the proportion of non-duplicate molecules among the generated samples. Novelty is assessed by comparing the SMILES representations [11] of the generated molecules with those in the training dataset.

Table S4: Effect of hybrid guidance weights ($w_1 = w_2 = w$) on structural validity and diversity metrics for QMe14S conditioned on μ . Both continuous and discrete guidance weights are varied simultaneously.

(w_1, w_2)	Validity	Mol. Stability	Uniqueness	Novelty
(1.0, 1.0)	0.993	0.974	0.961	0.845
(1.5, 1.5)	0.993	0.970	0.961	0.846
(2.0, 2.0)	0.991	0.967	0.958	0.842
(2.5, 2.5)	0.989	0.965	0.957	0.843
(3.0, 3.0)	0.987	0.963	0.956	0.846

Table S5: Effect of continuous-only guidance weight (w_1 , with $w_2 = 1.0$ fixed) on structural validity and diversity metrics for QMe14S conditioned on μ .

(w_1, w_2)	Validity	Mol. Stability	Uniqueness	Novelty
(1.0, 1.0)	0.993	0.974	0.961	0.845
(1.5, 1.0)	0.993	0.973	0.957	0.835
(2.0, 1.0)	0.993	0.975	0.956	0.839
(2.5, 1.0)	0.992	0.972	0.957	0.840
(3.0, 1.0)	0.993	0.972	0.950	0.836

Table S6: Effect of discrete-only guidance weight (w_2 , with $w_1 = 1.0$ fixed) on structural validity and diversity metrics for QMe14S conditioned on μ .

(w_1, w_2)	Validity	Mol. Stability	Uniqueness	Novelty
(1.0, 1.0)	0.993	0.974	0.961	0.845
(1.0, 1.5)	0.991	0.968	0.959	0.841
(1.0, 2.0)	0.987	0.966	0.960	0.842
(1.0, 2.5)	0.992	0.972	0.957	0.839
(1.0, 3.0)	0.993	0.972	0.950	0.836

S-5.5 Property Alignment

Full results of Property MAEs for six different properties and for the QMe14S data conditioned on dipole moment (μ) are shown in Table S7. The “Random” corresponds to MAEs between original molecular properties and fully shuffled properties, removing any correlations between structures and properties, hence serving as an upper bound. The “# Atoms” baseline uses atom counts as the predictor for molecular properties. The “QM9” baseline uses a separate predictor trained on a disjoint 50k molecules to predict properties of the 50k molecules used to train the generative model. The property predictor is trained on the QM9 xyz data [12], and provided by Hoogeboom *et al.* [9], and the corresponding MAE serves as a lower bound on achievable error. Improvement over the “# Atoms” baseline suggests that the generative model captures structural features beyond simple atom count when generating new molecules.

Table S7: Mean Absolute Error for molecular property prediction (lower is better). PropMolFlow (PMF) results employ Bayesian-optimized guidance weights. Top-ranked values are **bold**, second-best values are underlined. JODO results are from our own sampled molecules. ‘/’ indicates that the results are unavailable.

Property Units	α Bohr ³	$\Delta\epsilon$ meV	ϵ_{HOMO} meV	ϵ_{LUMO} meV	C_v cal/(mol·K)	μ -QM9 Debye	μ -QMe14S Debye
QM9 (Lower-Bound)	0.10	64	39	36	0.040	0.043	/
Random (Upper-Bound)	9.01	1470	645	1457	6.857	1.616	/
# Atoms	3.86	866	426	813	1.971	1.053	/
GeoLDM	2.37	587	340	522	1.025	1.108	/
GCDM	1.97	602	344	479	0.689	0.844	/
JODO	1.44	333	231	260	0.580	0.620	/
PMF-Vanilla	1.49	390	266	325	0.702	0.667	0.627
PMF-CFG	1.27	322	220	<u>265</u>	<u>0.581</u>	0.580	0.548
PMF-AG	<u>1.43</u>	344	242	274	0.638	0.631	<u>0.587</u>
PMF-MG	1.59	425	273	346	0.708	0.753	0.631

DFT confirms the property alignment predicted by a GVP property predictor on 500 molecules selected out of 10K generated molecules (Table S8).

S-5.6 Structural Validity

S-5.6.1 Molecule stability

Table S9 shows the molecule stability of generated molecules conditioning on six properties for baseline models and different guidance methods. To ensure chemically accurate stability evaluation

Table S8: Performance of property alignment for CFG, evaluated using both DFT and a property predictor (GVP). Metrics are computed on 500 molecules selected from the 10k generated molecules reported in Table S7, using the same property units. The 500 molecules were further filtered by molecule stability, closed-shell valence electron configuration, RDKit validity, PoseBusters validity, and DFT convergence.

Property	α	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	C_v	$\mu\text{-QM9}$	$\mu\text{-QMe14S}$
DFT vs Target	1.18	341	261	242	0.563	0.586	0.584
GVP vs Target	1.21	293	213	225	0.528	0.580	0.557

for the larger and more chemically diverse QMe14S dataset, we adopted the corrected valency framework proposed by Nikitin et al. [10].

Table S9: Performance of molecule stability (%). Higher numbers indicate better performance. All results for baseline models are from the PropMolFlow work [3]. The best results are in **bold** and the second best results are underlined.

Property	α	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	C_v	$\mu\text{-QM9}$	$\mu\text{-QMe14S}$
GeoLDM	81.4	83.1	84.0	84.0	81.3	85.5	/
GCDM	85.0	86.0	88.3	84.7	85.1	86.3	/
JODO	92.7	94.1	93.5	92.5	91.7	93.7	/
PMF							
Vanilla	92.8	94.6	95.1	<u>94.2</u>	91.8	96.2	97.9
CFG	93.1	92.5	95.6	92.0	88.4	92.8	95.8
AG	95.8	<u>95.7</u>	97.2	96.9	93.9	<u>96.6</u>	96.5
MG	<u>94.8</u>	96.9	<u>96.9</u>	93.3	<u>91.9</u>	96.8	<u>97.7</u>

S-5.6.2 RDKit validity

Table S10 shows the RDKit validity of generated molecules conditioning on six properties for baseline models and different guidance methods.

Table S10: RDKit validity (%) for generated molecules conditioned on six properties. All results for baseline models are from the PropMolFlow work [3]. The best results are in **bold** and the second best results are underlined.

Property	α	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	$\mu\text{-QM9}$	C_v	$\mu\text{-QMe14S}$
GeoLDM	91.6	91.8	92.2	92.2	93.0	89.6	/
GCDM	94.4	94.7	95.4	94.4	94.9	94.4	/
JODO	96.4	97.0	95.9	95.6	96.4	95.6	/
PMF							
Vanilla	97.6	98.4	98.3	<u>98.2</u>	98.7	<u>97.0</u>	99.0
CFG	96.8	97.4	97.9	96.6	96.5	94.0	98.8
AG	98.2	97.3	<u>98.5</u>	98.0	98.0	97.6	99.3
MG	<u>97.6</u>	<u>98.2</u>	98.9	96.3	<u>98.6</u>	96.8	<u>99.2</u>

S-5.6.3 PoseBusters validity

Table S11 shows the PoseBusters validity of generated molecules conditioning on six properties for baseline models and different guidance methods.

Table S11: PoseBusters validity (%) for generated molecules conditioned on six properties. All results for baseline models are from the PropMolFlow work [3]. The best results are in **bond** and the second best results are underlined.

Property	α	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	$\mu\text{-QM9}$	C_v	$\mu\text{-QMe14S}$
GeoLDM	89.1	89.2	90.3	89.9	90.3	87.3	/
GCDM	91.6	92.0	92.9	92.0	92.4	91.4	/
JODO	95.2	95.7	94.7	94.5	95.3	94.1	/
PMF							
Vanilla	95.7	97.3	96.6	96.5	<u>97.1</u>	<u>95.5</u>	99.0
CFG	95.4	95.3	96.5	94.1	92.9	90.7	96.2
AG	96.7	94.6	<u>97.3</u>	97.5	94.0	96.2	98.4
MG	<u>95.9</u>	<u>97.1</u>	97.5	95.2	97.5	<u>95.5</u>	<u>98.9</u>

S-5.7 Structural Diversity

Table S12 shows that percentages of molecules that are RDKit valid and unique in their SMILES representation.

Table S12: Ratios of unique RDKit valid molecules ('Uniqueness') across different guidance methods. All values are reported in unit of '%', and higher numbers indicate higher structural diversity. The highest values are in **bond**, and the second highest values are underlined.

Property	α	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	C_v	$\mu\text{-QM9}$	$\mu\text{-QMe14S}$
Vanilla	96.0	96.6	<u>96.5</u>	95.6	95.6	96.5	94.9
CFG	94.3	95.3	95.4	94.5	92.3	95.0	94.9
AG	95.7	93.5	95.6	94.8	95.2	94.7	94.0
MG	<u>95.9</u>	<u>96.2</u>	96.6	<u>95.0</u>	95.6	<u>96.4</u>	94.9

Table S13 shows that all guidance methods increase bond entropy for most properties compared to the vanilla models. Among guidance approaches, CFG achieves the highest bond entropy for $\Delta\epsilon$, ϵ_{HOMO} and μ , whereas AG has the highest bond entropy for α and C_v . This elevated bond entropy under CFG and AG can likely be attributed to its relatively high guidance weights on atomic positions (Table S1).

S-5.7.1 Bond-order entropy

Table S13: Bond-order entropy of generated molecules under various guidance methods. The highest values are in **bold**, and the second highest values are underlined.

Property	α	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	$\mu\text{-QM9}$	C_v	$\mu\text{-QMe14S}$
Vanilla	0.593	0.560	0.543	0.495	0.480	0.490	0.457
CFG	0.65	0.569	0.532	<u>0.562</u>	0.536	0.617	0.503
AG	0.599	0.657	0.598	0.556	0.556	0.498	0.598
MG	0.524	<u>0.570</u>	<u>0.578</u>	0.567	0.463	<u>0.510</u>	0.433

S-5.7.2 Element entropy

Table S14: Element entropy of generated molecules under various guidance methods. The highest values are in **bold**, and the second highest values are underlined.

Property	α	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	$\mu\text{-QM9}$	C_v	$\mu\text{-QMe14S}$
Vanilla	1.570	1.538	1.538	1.514	1.526	1.550	1.698
CFG	1.570	<u>1.540</u>	<u>1.545</u>	1.551	1.557	1.577	1.770
AG	<u>1.568</u>	1.551	1.569	<u>1.524</u>	<u>1.545</u>	<u>1.569</u>	<u>1.730</u>
MG	1.510	1.533	1.509	1.523	1.530	1.523	1.709

S-5.7.3 Scaffold diversity

Table S15: Scaffold diversity of generated molecules under various guidance methods. The highest values are in **bold**, and the second highest values are underlined.

Property	α	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	$\mu\text{-QM9}$	C_v	$\mu\text{-QMe14S}$
Vanilla	0.272	0.325	0.330	0.305	0.496	0.279	0.216
CFG	<u>0.284</u>	<u>0.314</u>	0.317	<u>0.310</u>	<u>0.403</u>	0.331	0.309
AG	0.247	0.294	0.289	0.302	0.270	<u>0.281</u>	0.228
MG	0.369	0.242	<u>0.328</u>	0.354	0.331	0.277	<u>0.281</u>

S-5.7.4 Novelty

Table S16: Novelty (%) of generated molecules under various guidance methods, computed as the fraction of generated molecules whose canonical SMILES are absent from the training set. The highest values are in **bold**, and the second highest values are underlined.

Property	α	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	$\mu\text{-QM9}$	C_v	$\mu\text{-QMe14S}$
Vanilla	<u>95.3</u>	<u>95.2</u>	95.2	95.6	<u>96.5</u>	<u>97.0</u>	<u>84.2</u>
CFG	95.0	95.6	<u>95.0</u>	<u>96.3</u>	96.9	96.7	84.6
AG	94.7	94.5	<u>95.0</u>	95.5	95.1	97.2	82.6
MG	96.8	94.8	94.2	96.4	95.6	95.9	82.6

S-5.8 Training time cost

Table S17 reports the training and sampling wall-clock time for each guidance method. Both training and sampling used one NVIDIA A100 GPU with 80GB memory. Because AG uses the PMF-vanilla’s model as its main model, its total training time for the main model along is unchanged. In practice, one needs to either save a checkpoint model with fewer training epochs as the guide model or to train model with less complex architecture (for example, fewer node and edge features), which only takes a negligible additional 2 hours. CFG slightly reduces training time by skipping the property-embedding MLP operations on 10% of the data for the unconditional model. MG incurs a modest training overhead relative to PMF-Vanilla because it maintains an EMA copy of the online model. The MG model thus requires two forwards to obtain its unconditional and conditional predictions to compile the updated training objective (Eq. 10) without gradient computation, and the EMA copy is updated every training step. The discussion for sampling efficiency can be found in Figure 6.

Table S17: Training and sampling times. Training is for 2000 epochs and sampling is for 10 k molecules. Values are reported as mean $\pm$ standard deviation across the six properties.

Model	Training [h] $\downarrow$	Sampling [min] $\downarrow$
Vanilla	55.4 $\pm$ 0.7	9.6 $\pm$ 0.5
CFG	54.9 $\pm$ 0.5	16.8 $\pm$ 0.4
AG	57.4 $\pm$ 0.7	12.4 $\pm$ 1.3
MG	90.3 $\pm$ 3.2	9.1 $\pm$ 0.5

S-5.9 Ablations

Guidance weights. Figure S4 shows how molecule stability varies with guidance weights. For both AG and CFG, increasing the weight on atomic positions (w_1) while decreasing the weight on discrete variables (w_2) enhances molecule stability. However, AG is far more sensitive to changes in w_2 than CFG: AG’s molecule stability plummets from over 0.9 at $w_2 = 1$ to about 0.1 at $w_2 = 3$, whereas CFG experiences a much smaller drop (≤ 0.15). This suggests that CFG models are more robust to choices of discrete-variable weights, likely because CFG uses a single network with alternate property embeddings for unconditional and conditional generation, while AG relies on two more loosely coupled networks whose logits of discrete-variable predictions may not be compatible when interpolated with large weights.

The guidance-weight variation for property alignment closely mirror the trends observed in our Bayesian optimization experiments (see Figure S5). Nevertheless, MG models exhibit almost no dependence of molecule stability, bond entropy and property alignment on guidance weights (Table S20), suggesting that MG struggles to leverage the guidance scaling effects—a phenomenon that merits further investigation.

Inference timesteps for CFG. For a good balance between sampling efficiency and accuracy, we use 100 integration timesteps to generate molecules. Table S18 reports the property MAEs for CFG inference with varying numbers of time steps (n_{ts}). Raising n_{ts} to 200 causes all MAEs to be lower than those of JODO (Table S7). Further increasing n_{ts} improves the alignment for $\Delta\epsilon$, ϵ_{HOMO} , and ϵ_{LUMO} , but has negligible impact on other properties.

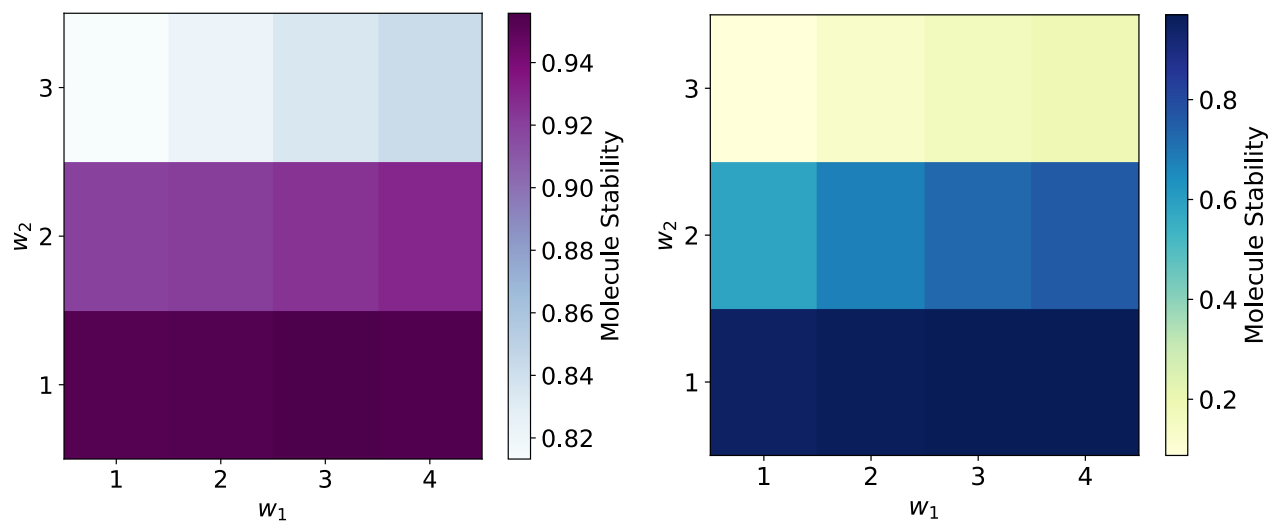


Figure S4: Dependence of molecule stability on guidance weights for CFG (Left) and AG (Right). Values are reported by averaging across six properties.

Table S18: MAEs for CFG with optimized guidance weights across varying integration timesteps (n_{ts}). Best results are bolded.

n_{ts}	α	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	$\mu\text{-QM9}$	C_v	$\mu\text{-QMe14S}$
100	1.27	322	220	265	0.580	0.581	0.557
200	1.24	317	222	252	0.552	0.568	0.541
300	1.22	313	219	248	0.559	0.573	0.533
400	1.22	311	219	253	0.564	0.575	0.541
500	1.23	309	215	246	0.559	0.574	0.532

Guidance on all four molecular modalities. Table S19 compares the property MAEs for CFG using Bayesian-optimized guidance weights across all four modalities, namely atom types, formal charges, bond orders and atom coordinates. The results show that using four separate guidance weights yields performance comparable to using just two: one for positions and one shared across all discrete variables. Similar results are observed for AG (Table S21).

Table S19: Comparison of property MAEs with four guidance weights against two guidance weights for CFG.

CFG	α	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	$\mu\text{-QM9}$	C_v	$\mu\text{-QMe14S}$
Two weights	1.27	322	220	265	0.580	0.581	0.557
Four weights	1.25	343	219	270	0.571	0.591	0.552

Figure S5 shows the property alignment as a function of guidance weights w_1 and w_2 . The guidance weights for the lowest MAE of CFG ϵ_{HOMO} are $(w_1, w_2) = (4, 2)$ with a property MAE of 216 meV, and the second lowest MAE comes with $(w_1, w_2) = (3, 2)$ with a MAE of 221 meV, which confirms the findings of the Bayesian analysis. The guidance weights for the lowest MAE of AG C_v are $(w_1, w_2) = (3, 1)$ with an MAE of 0.654 cal/(mol·K), which is slightly worse than the result (0.638 cal/(mol·K)) using the Bayesian-optimized guidance weights $(w_1, w_2) = (2.75, 1.14)$.

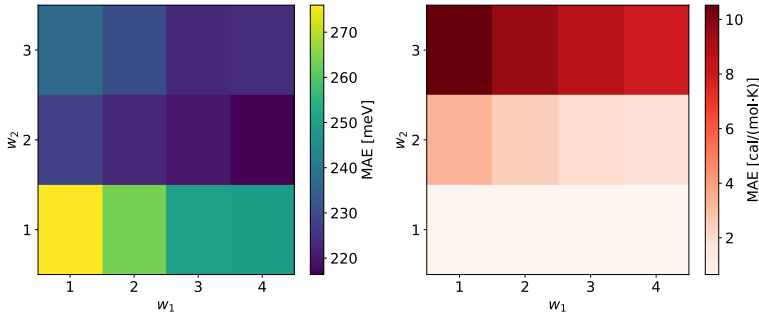


Figure S5: Effects of guidance weights on the property alignment for CFG ϵ_{HOMO} (Left) and AG C_v (Right).

Ablation study on the dependence of molecule stability on guidance weights for MG can be found in Table S20.

Table S20: Molecule Stability for MG versus guidance weights. This molecule stability and bond distance std is averaged across models conditioned on six properties.

w	Molecule Stability [%]	Bond Entropy	MAE, $\Delta\epsilon$ [meV]
1	95.1 ± 2.1	0.536 ± 0.044	429
2	95.0 ± 2.1	0.535 ± 0.044	427
3	95.1 ± 2.3	0.533 ± 0.044	429
4	95.1 ± 2.1	0.536 ± 0.048	429

Results for guidance on four weights versus guidance on two weights for AG can be found in Table S21.

Table S21: Comparison of property MAEs with four guidance weights against two guidance weights for AG.

AG	α	$\Delta\epsilon$	ϵ_{HOMO}	ϵ_{LUMO}	$\mu\text{-QM9}$	C_v	$\mu\text{-QMe14S}$
Two weights	1.43	344	242	274	0.631	0.638	0.595
Four weights	1.46	341	247	265	0.620	0.640	0.584

S-6 Original QM9 Properties and Data Details

Exact definitions for all six molecular properties of QM9 data are shown below.

- α (Polarizability): Tendency of a molecule to acquire an electric dipole moment when subjected to an external electric field.
- $\Delta\epsilon$: The energy gap between HOMO and LUMO.
- ϵ_{HOMO} : Highest occupied molecule orbital energy.
- ϵ_{LUMO} : Lowest unoccupied molecule orbital energy.
- μ : Dipole moment, which measures the separation of positive and negative charges within a molecule.
- C_v : Heat capacity at room temperature 298.15 K.

QM9 is a 134k small molecule dataset that only contains of up to 9 heavy atoms (C, N, O, F). The atom sizes range from 3 to 29 with an average of 18 atoms, including explicit hydrogen. All molecules are optimized by density functional theory (DFT) calculations and thus in their stable states. By design, all molecules in QM9 are charge neutral and have a close shell valence electron configuration. But one should note there are molecules that carry explicit atom formal charges and a molecule graph carrying this information might be hence helpful—like we did in this work—to generate molecules with valid charge–valency configuration.

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